

Statistics and Probability Notes

DEPARTMENT OF MATHEMATICS & DATA SCIENCE

Course: Probability Theory and Mathematical Statistics | Unit 3: Limit Theorems

1. Law of Large Numbers and Central Limit Theorem:

The Central Limit Theorem provides the foundation for statistical estimation. When drawing independent random samples from any distribution with finite variance, the standardized sum converges in distribution to a standard normal distribution.

2. Standard Error of the Mean:

The standard error equals σ divided by the square root of n . As sample size grows, the standard error shrinks, meaning our sample mean concentrates around the population parameter.

3. Applications in Statistical Inference:

Z-tests, t-tests, and regression analysis rely on the normality assumption guaranteed asymptotically by the Central Limit Theorem.